A. Replacement reactions involving loss of nitrogen

1. Replacement by hydrogen

Benzene diazonium chloride on reduction with mild reducing agents like hypophosphrous acid (phosphinic acid) or ethanol in the presence of cuprous ion gives benzene. This reaction proceeds through a free-radical chain mechanism.

$$C_6H_5\text{-}\overset{+}{N}_2\,Cl^- + H_3PO_2 + H_2O \xrightarrow{CuCl} C_6H_6 + H_3PO_3 + HCl + N_2\uparrow$$

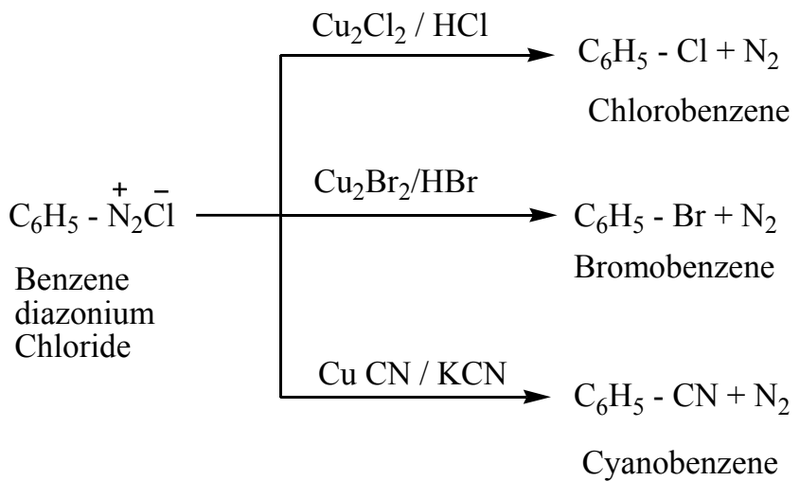
$$C_6H_5\text{-}\overset{+}{N}_2\,Cl^- + CH_3CH_2OH \rightarrow C_6H_6 + N_2 + CH_3CHO + HCl$$

2. Replacement by Chlorine, Bromine, Cyanide group

a) Sandmeyer reaction

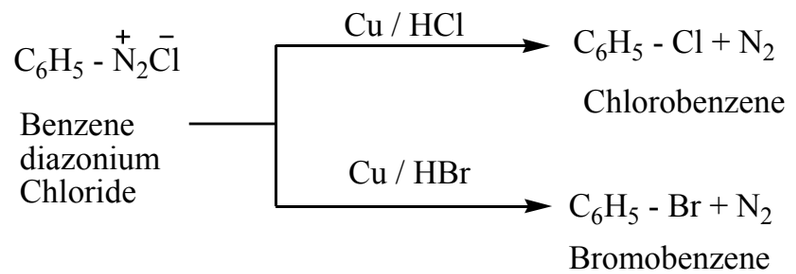
On mixing freshly prepared solution of benzene diazonium chloride with cuprous halides (chlorides and bromides), aryl halides are obtained. This reaction is called Sandmeyer reaction.

When diazonium salts are treated with cuprous cyanide, cyanobenzene is obtained.



b) Gattermann reaction

Conversion of benzene diazonium chloride into chloro / bromo arenes can also be effected using hydrochloric / hydrobromic acid and copper powder. This reaction is called Gattermann reaction.



The yield in Sandmeyer reaction is found to be better than the Gattermann reaction.

3. Replacement by iodine

Aqueous solution of benzene diazonium chloride is warmed with KI to form iodobenzene

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